

# Mohit Saini

## Senior Software Developer

### Skills

- Python
- Git Version Control
- Machine Learning
- Deep Learning
- PyTorch
- AWS
- MCP
- LLM
- NumPy
- Model Optimization
- AI Ops
- LangChain
- Scikit-learn
- Pandas
- AI Agent

### Professional Experience

Senior Software Developer  
Software Developer  
Associate Software Developer

*March 2025 - Present*  
*February 2023 - February 2025*  
*January 2021 - January 2023*

### Projects

#### RAG Optimization & Evaluation System

- Developed a scalable RAG optimization and evaluation system to enhance retrieval quality, ensure reliable LLM responses, and establish performance benchmarks for enterprise-grade AI applications.
- Designed and implemented an advanced Retrieval-Augmented Generation (RAG) optimization framework to improve retrieval quality and response accuracy for enterprise AI applications
- Developed and evaluated multiple chunking strategies including fixed-size, semantic, and hierarchical chunking to enhance contextual retrieval performance
- Established evaluation frameworks using metrics such to benchmark retrieval performance
- Implemented guardrails including PII detection, toxicity filtering, hallucination reduction, and prompt injection protection
- Developed automated pipelines for continuous evaluation and monitoring of RAG system performance
- Deployed scalable evaluation and retrieval services using GCP (Vertex AI, BigQuery, Cloud Run)
- Collaborated with engineering and product teams to improve retrieval strategies and align AI performance with business goals
- Documented evaluation methodologies, system architecture, and performance benchmarks for continuous improvement

#### Vertex AI-Based GenAI Platform

- An enterprise initiative to build a production-grade Generative AI platform using Vertex AI, enabling scalable deployment, monitoring, and optimization of LLM-based applications across multiple business use cases.
- Designed and deployed Generative AI solutions using Vertex AI including Workbench, Model Registry, and Pipelines for end-to-end model lifecycle management

- Built scalable MLOps workflows covering data ingestion, model training, evaluation, deployment, and monitoring
- Integrated BigQuery and Dataflow pipelines for large-scale data processing and real-time data ingestion
- Developed LLM-powered applications including chatbots, document summarization, and knowledge-based question answering systems
- Deployed models using docker ensuring high availability, scalability, and fault tolerance
- Implemented feature store integration for consistent and reusable machine learning features across pipelines
- Automated CI/CD pipelines for continuous integration and deployment of AI models
- Monitored model performance and implemented logging, alerting, and observability for production systems
- Collaborated with cross-functional teams to align AI solutions with business requirements and ensure seamless deployment
- Documented system architecture, workflows, and best practices for maintainability and scalability

## AI Lead Generation & Outreach Automation

- Built an end-to-end automated sales funnel replacing a fully manual outreach process, covering lead
- discovery, AI qualification, personalised proposal generation, and outreach delivery as a daily automated pipeline
- Developed Python-based web scrapers using Playwright to extract active reseller profiles, listing data, pricing patterns, and seller activity metrics from Depop and Vinted at scale, handling dynamic rendering, pagination, and anti-bot resilience.
- Designed ETL pipelines to ingest, clean, and normalize scraped seller data into BigQuery, standardizing inconsistent listing formats, deduplicating profiles, and structuring data for downstream AI processing.
- Built an AI qualification engine using Claude to score and filter scraped leads against the client's ideal customer profile, evaluating inventory volume, category alignment, pricing tier, and activity frequency.
- Engineered a dynamic proposal generation system using Claude and prompt engineering, comparing each qualified seller's active listings against the client's wholesale catalogue to generate personalized bundle proposals.
- Orchestrated the full pipeline using n8n, scheduling daily scrape runs, triggering AI qualification workflows, routing qualified leads into outreach sequences, and managing webhook-based event flows between stages

## Customer Behavior Analysis using Deep Learning

- Built an AI-driven customer behavior analysis system to understand user patterns, predict customer actions, and improve business decision-making using Deep Learning and advanced data analytics.
- Extracted and processed large customer datasets from multiple sources using Python, SQL, and data preprocessing techniques.
- Developed Deep Learning models using TensorFlow and PyTorch to analyze user activity, purchasing patterns, and engagement behavior.
- Applied clustering, classification, and regression algorithms to segment customers and predict future actions.
- Used feature engineering, normalization, and dimensionality reduction techniques to improve model performance.
- Designed data pipelines for training, validation, and deployment of Machine Learning models.
- Built analytical dashboards to visualize predictions, customer segments, and performance metrics.

- Worked with business teams to integrate model outputs into marketing and recommendation strategies.

## **AI-Based Recommendation Engine with Custom Machine Learning Models**

- Developed a scalable AI-based recommendation engine for a large e-commerce platform to provide personalized product recommendations using custom Machine Learning algorithms. The objective was to design intelligent models capable of learning user behavior, predicting preferences, and optimizing recommendations without relying on third-party AI APIs.
- Collected and processed large-scale user interaction data, including clicks, purchases, search history, and browsing patterns using Python and SQL to build structured datasets for Machine Learning training.
- Designed and implemented recommendation algorithms including collaborative filtering, content-based filtering, and hybrid models using NumPy, Pandas, and Scikit-learn.
- Developed custom Machine Learning models to predict user preferences using classification, regression, and similarity-based techniques.
- Applied mathematical concepts such as vector similarity, matrix factorization, probability, and optimization algorithms to improve recommendation accuracy.
- Built data pipelines for data preprocessing, feature engineering, model training, and evaluation to support continuous learning.
- Implemented model evaluation metrics such as precision, recall, MAP, and RMSE to measure recommendation quality and performance.
- Designed backend architecture to support real-time recommendation generation and scalable deployment.
- Optimized model performance using hyperparameter tuning and cross-validation to ensure high accuracy and low latency.
- Collaborated with product and engineering teams to integrate the recommendation engine into the production system and align results with business objectives.